

Session 5 – Thinking London Spatially Further (Spatial Query with SpatiaLite Advance)

Objectives: To develop skills of spatial query with SQL in SpatiaLite. Explore a range of spatial variables of London.

Connect the existing SQLite Database “test.sqlite”

Task 1: Spatial Extent of London

Union multiple conditions

Using spatial functions of “Area”, “Mbr” along with “Max”, “Min”

Find out London boroughs with the smallest / largest area, with the lowest / highest population density, as the southernmost / northernmost / most east / most west borough.

Task 2: Neighbour Boroughs

Join tables with multiple conditions

Using spatial relation of “Touches”

Find out adjacent London boroughs.

Task 3: An Isolated Island in London?

Insert a new row as the isolated island

Left join neighbour borough to make sure that all London boroughs included

Using spatial relation of “Disjoint”

Using post-processing of “Having” to reduce the result-set of query

Find out isolated district.

Task 4: Business / Services within Boroughs (Points within Polygons)

Open Visual Shapefile “credit_union.shp”

Join or Left Join the Point table and the Polygon table

Using spatial relation of “Contain”

Find out which credit unions within which London boroughs.

Try the same procedure with the table of “UndergroundStations”

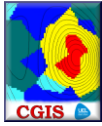
Find out which underground stations within which London boroughs.

Task 5: Near Neighbour Business / Services (Distance between Points)

(Need to load Shapefile “credit_union.shp”)

Using spatial relation of “Distance” calculated by “Length”.

Find out all credit Unions within distance < 2000 meters for each credit union.



Task 6: Motorways across London Boroughs (Intersection)

Open Shapefile “motorway.shp”

Using spatial relation of “Intersect”

Join the other table

Find out Which motorways across which London boroughs.

Task7: Tube Stations near Motorways in London (Distance between Points and Lines)

(Need to load Shapefile “UndergroundStations.shp”)

Using spatial relation of “Distance”

Multiple Join and Left Join

Find out distance between Underground Stations and Motorways.

Task 8: Buffers of Motorways

Create a table of motorway_buffers with Multipolygon geometry

Using spatial function of “CastToMultipolygon” to create 1km buffer zones for Motorways

Using spatial function of “CastToMultipolygon” to create 1km – 2.5km buffer rings for Motorways.

Task 9: Merging London Boroughs

Create a view of view_london

Using spatial relation of “Union” along with Group to merge London boroughs by Inner / Outer borough

Further merge all London boroughs into one polygon in different way.

Task 10: Add SpatiaLite Table(s) into GIS

Show maps in QGIS.

Disconnect the current SQLite Database

Please export the following CSV files and put them into your portfolio: the query of london_guinness_book in **Task1**, the count of underground station by Borough in **Task4**, motorways across London in **Task 6**, and underground stations within 1 km of motorways in **Task 7**.

Please export maps (in PNG format from SpatiaLite, or in other formats from QGIS) from **Task 8**, and put them into your portfolio.

Please have comment on each Task (such as what other applications can use these SQL commands / functionalities, could these SQL commands could be extended / improved / simplified ...).